

CENTAFLEX-A

HIGHLY FLEXIBLE & MULTI-DIRECTIONAL ELASTIC COUPLINGS

Product Overview: The CENTAFLEX-A series represents the industry standard for highly flexible, multi-directional elastic couplings. Engineered with a segmented precompressed rubber element design, it excels at dampening severe torsional oscillations, tuning natural resonances, and neutralizing parallel, axial, and angular misalignments across complex industrial drive chains.

KEY PERFORMANCE ADVANTAGES

- Precompressed Rubber Design:** The specialized element geometry maintains exceptional durability under extreme compression vectors, extending service life significantly.
- Dynamic Tuning Flexibility:** Available in variable rubber qualities (50 Shore A and 60 Shore A) to isolate critical system natural frequencies cleanly.
- Versatile Application Spectrum:** Extensively specified for main pump packages, generator assemblies, marine propulsion lines, and heavy compressor modules.

TECHNICAL SPECIFICATIONS & PERFORMANCE DATA (SIZES 30 - 800)

Below is the calibrated design matrix for the **CENTAFLEX-A Sizes 30 to 800**, featuring robust dimensions required for heavy elastic industrial coupling setups.

Size	Rubber Quality [Shore A]	Nominal Torque	Maximum Torque	Continuous Vibratory	Permissible Power Loss	Dynamic Torsional Stiffness	Relative Damping	Max Speed	Axial Displacement		Radial Displacement		Angular Displacement	
		T_{KN} [kNm]	T_{Kmax} [kNm]	T_{KW} [kNm]	P_{KV} [W]	C_{Tdyn} [kNm/rad]	ψ	n_{max} [min ⁻¹]	ΔK_a [mm]	C_a [kN/mm]	ΔK_r [mm]	C_r [kN/mm]	ΔK_α [°]	C_α [kNm/°]
30	50	0.500	1.400	0.200	80	4.80	0.80	4000	5	0.2250	2	0.500	3	0.01800
	60	0.500	1.400	0.200	80	7.80	1.00			0.4600		1.000		0.02500
50	50	0.700	2.100	0.300	90	12.00	0.80	4000	5	0.4750	2	1.450	2	0.03000
	60	0.700	2.100	0.300	90	19.00	1.00			0.8500		2.350		0.04000
80	50	0.900	2.100	0.320	100	16.00	0.80	4000	3	0.4500	1.5	1.600	2	0.03800
	60	0.900	2.100	0.320	100	25.00	1.00			0.7500		2.400		0.05000
90	50	1.100	3.150	0.450	120	10.50	0.80	3600	5	0.3150	2	0.800	3	0.02800
	60	1.100	3.150	0.450	120	16.00	1.00			0.6500		1.500		0.03500
140	50	1.700	4.900	0.700	150	26.50	0.80	3600	5	0.5800	2	1.500	2	0.04200
	60	1.700	4.900	0.700	150	40.00	1.00			1.0000		2.500		0.06700
200	50	2.400	6.000	0.960	170	38.70	1.05	3000	5	0.6400	2	1.500	2	0.06200
	60	2.400	6.000	0.960	170	60.00	1.10			1.1000		2.650		0.06800
250	50	3.000	8.750	1.250	200	43.00	0.80	3000	5	0.6600	2	1.700	2	0.07100
	60	3.000	8.750	1.250	200	77.00	1.00			1.2000		2.700		0.10500

Size	Rubber Quality [Shore A]	Nominal Torque	Maximum Torque	Continuous Vibratory	Permissible Power Loss	Dynamic Torsional Stiffness	Relative Damping	Max Speed	Axial Displacement		Radial Displacement		Angular Displacement	
		T_{KN} [kNm]	T_{Kmax} [kNm]	T_{KW} [kNm]	P_{KW} [W]	C_{Tdbm} [kNm/rad]	ψ	n_{max} [min ⁻¹]	ΔK_a [mm]	C_a [kN/mm]	ΔK_r [mm]	C_r [kN/mm]	ΔK_w [°]	C_w [kNm/°]
400	50	5.000	12.500	2.000	240	75.00	1.05	2500	5	0.7000	2	1.900	2	0.09500
	60	5.000	12.500	2.000	240	120.00	1.10			1.3500		3.000		0.13500
600	50	8.000	20.000	3.200	330	105.00	1.05	2500	5	0.8500	2	2.500	2	0.13000
	60	8.000	20.000	3.200	330	160.00	1.10			1.5500		3.700		0.20000
800	50	12.500	30.000	5.000	420	160.00	1.05	2300	5	1.1000	2	3.300	2	0.43000
	60	12.500	30.000	5.000	420	243.00	1.10			2.0000		5.000		0.56000

* Values for 70 and 75 Shore A rubber quality variants are readily available on request.

** Speed ratings specified at 60 Shore A rubber quality values.

PROCUREMENT, SHORE SELECTION & VIBRATION ANALYSIS SUPPORT (INDIA)

For dynamic torsional vibration analysis (TVA), custom 70/75 Shore element requests, or rapid quote processing, please contact our expert engineering desk:

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